

\* Rank 0 tensor. \* Rank 1 tensor. \* Rank 2 tensor (s). \* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ -G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = Q_{pz}^{(1)}$$

\* Rank 3 tensor (ss).

$$\begin{bmatrix} G_{xxx} & 0 & 0 \\ -G_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -G_{xxx} & 0 \end{bmatrix}$$

$$G_{xxx} = G_{f2}^{(1)}$$

\* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & -G_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzx} = 3Q_{du}^{(1)}$$

\* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = -Q_{du}^{(2)} + Q_s^{(1)}$$

$$G_{xyz} = 2Q_{du}^{(2)} + Q_s^{(1)}$$

\* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & -G_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -G_{xxxx} \\ G_{xxxx} & -G_{xxxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -G_{xxxx} & 0 & 0 \end{bmatrix}$$

$$G_{xxxx} = G_{ga}^{(1)}$$

\* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{xxxx} & G_{xxxy} \\ 0 & 0 & 0 & 0 & -G_{xxxx} & -G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{yzxz} & G_{xxzx} \\ -G_{xxxx} & G_{xxxx} & 0 & -G_{yzxz} & 0 & 0 \\ -G_{xxxy} & G_{xxxy} & 0 & -G_{xxxx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= -2Q_{f1}^{(1)} \\
G_{xxxy} &= -2Q_{faz}^{(1)} + 2Q_{pz}^{(1)} \\
G_{yyzz} &= -4Q_{faz}^{(1)} - Q_{pz}^{(1)}
\end{aligned}$$

\* Rank 4 tensor (aas). \* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yyzz} & 0 \\ -G_{yyzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yyzz} = -Q_{pz}^{(2)}$$

\* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxxx} & G_{xxxy} \\ 0 & -G_{xxxx} & G_{xxxy} \\ 0 & 0 & G_{zzxy} \\ 0 & G_{yyzz} & 0 \\ G_{yyzz} & 0 & 0 \\ G_{xxxx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= Q_{f1}^{(2)} \\
G_{xxxy} &= -Q_{faz}^{(2)} + Q_{pz}^{(3)} \\
G_{zzxy} &= 2Q_{faz}^{(2)} + Q_{pz}^{(3)} + 2Q_{pz}^{(4)} \\
G_{yyzz} &= -Q_{faz}^{(2)} + Q_{pz}^{(4)}
\end{aligned}$$

\* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yyzz} & G_{yzxy} \\ G_{yzxy} & -G_{yzxy} & 0 & G_{yyzz} & 0 & 0 \\ G_{xyxx} & G_{xyxx} & G_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{yyzz} &= -Q_{faz}^{(3)} + Q_{pz}^{(6)} \\
G_{yzxy} &= Q_{f1}^{(3)} \\
G_{xyxx} &= -Q_{faz}^{(3)} + Q_{pz}^{(5)} \\
G_{xyzz} &= 2Q_{faz}^{(3)} + Q_{pz}^{(5)} + 2Q_{pz}^{(6)}
\end{aligned}$$

\* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{xxxx} & G_{xxxy} & 0 & G_{xxxz} & G_{xxyx} \\ 0 & 0 & 0 & 0 & -G_{xxxx} & -G_{xxyx} & 0 & -G_{xxxz} & -G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & -G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yyzz} & G_{yzxy} & 0 & G_{yyzz} & G_{yzxy} \\ -G_{yzxy} & G_{yzxy} & 0 & -G_{yyzz} & 0 & 0 & -G_{yyzz} & 0 & 0 \\ -\frac{G_{xxxxy}}{2} - \frac{G_{xxxyx}}{2} & \frac{G_{xxxxy}}{2} + \frac{G_{xxxyx}}{2} & 0 & -G_{xxxz} & 0 & 0 & -G_{xxxz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= G_{ga}^{(1)} - 2Q_{f1}^{(1)} + Q_{f1}^{(2)} \\
G_{xxxy} &= -2Q_{faz}^{(1)} - Q_{faz}^{(2)} + 2Q_{pz}^{(1)} + Q_{pz}^{(3)} \\
G_{xxxz} &= G_{ga}^{(1)} - 2Q_{f1}^{(1)} - Q_{f1}^{(2)} \\
G_{xxyx} &= -2Q_{faz}^{(1)} + Q_{faz}^{(2)} + 2Q_{pz}^{(1)} - Q_{pz}^{(3)} \\
G_{zzxy} &= 2Q_{faz}^{(2)} + Q_{pz}^{(3)} + 2Q_{pz}^{(4)} \\
G_{yyzz} &= -4Q_{faz}^{(1)} - Q_{faz}^{(2)} - Q_{pz}^{(1)} + Q_{pz}^{(4)} \\
G_{yzxy} &= -G_{ga}^{(1)} - 2Q_{f1}^{(1)} \\
G_{yyzz} &= -4Q_{faz}^{(1)} + Q_{faz}^{(2)} - Q_{pz}^{(1)} - Q_{pz}^{(4)}
\end{aligned}$$

\* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & G_{yzxy} & 0 & G_{yzxz} & G_{yzxy} \\ G_{yzxy} & -G_{yzxy} & 0 & G_{yzxz} & 0 & 0 & G_{yzxz} & 0 & 0 \\ G_{xyxx} & G_{xyxx} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = -Q_{faz}^{(3)} - Q_{pz}^{(2)} + Q_{pz}^{(6)}$$

$$G_{yzxy} = Q_{f1}^{(3)}$$

$$G_{yzxz} = -Q_{faz}^{(3)} + Q_{pz}^{(2)} + Q_{pz}^{(6)}$$

$$G_{xyxx} = -Q_{faz}^{(3)} + Q_{pz}^{(5)}$$

$$G_{xyzz} = 2Q_{faz}^{(3)} + Q_{pz}^{(5)} + 2Q_{pz}^{(6)}$$

\* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & G_{xxxx} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -G_{xxxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{xxxx} & 0 & 0 \\ 0 & 0 & -G_{xxxx} \\ 0 & -G_{xxxx} & 0 \end{bmatrix}$$

$$G_{xxxx} = G_{ga}^{(1)}$$